

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I & PHASE – II

Analysis and Architectural Modelling (CO1 & CO2)

Group No: B1

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PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

1. Problem Title: Personal Task Manager System

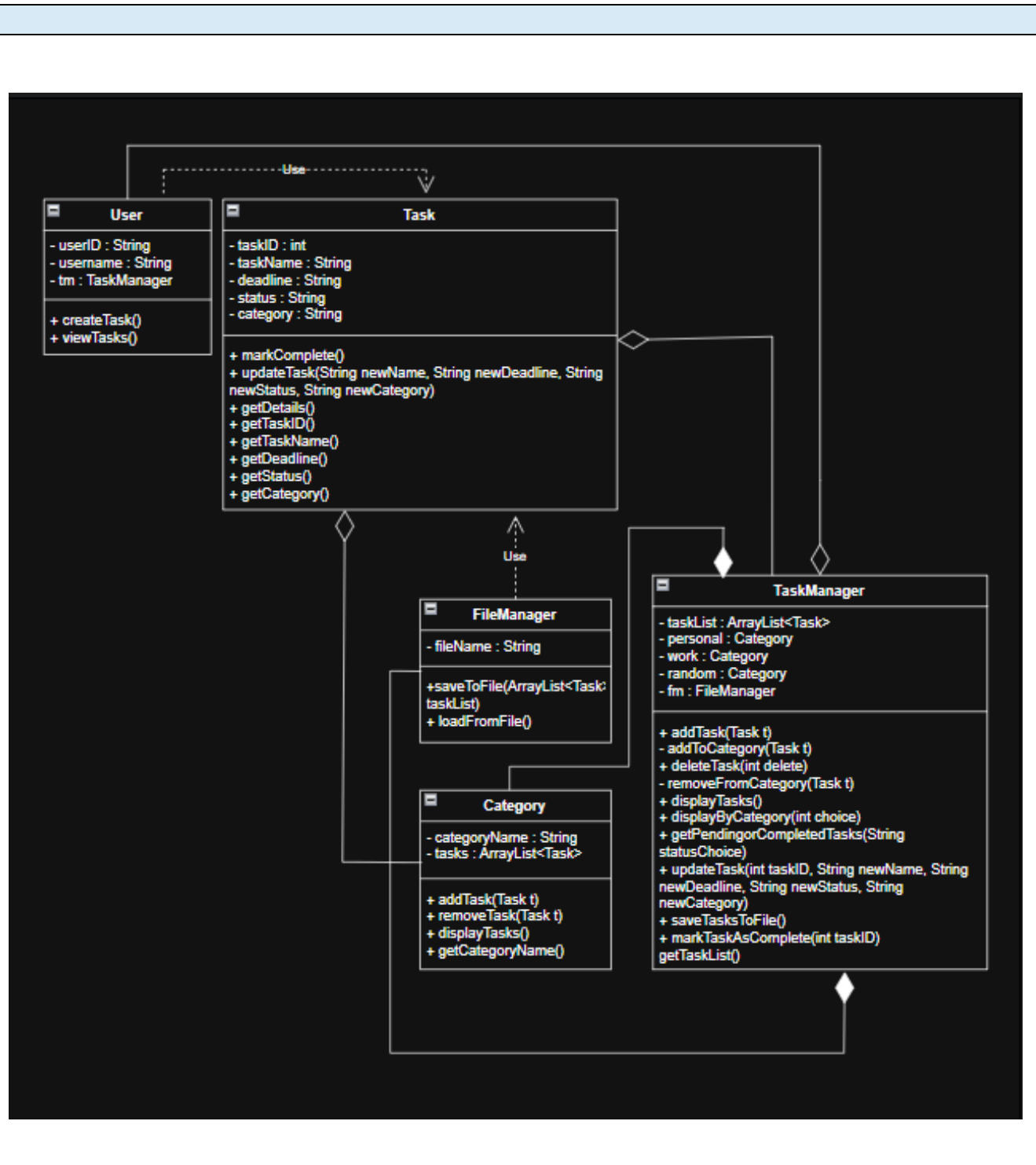
2. Problem Description: A personal task manager helps users organize daily activities, track pending tasks, and manage deadlines. The system should allow users to add tasks, update task status, and view task lists.

3. Identified Entities (Classes)

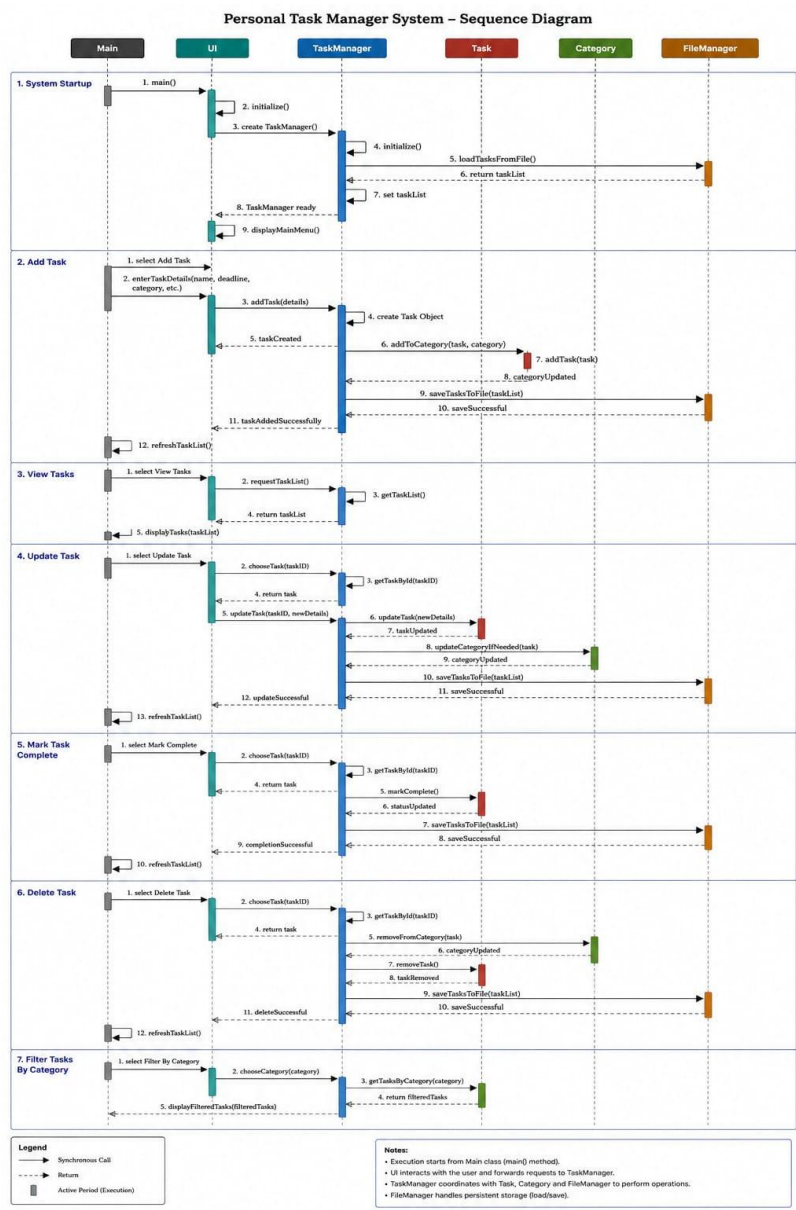
Class Name	Description
User	Stores user details and interacts with the system.
Task	Stores task information like name, deadline, and status.
TaskManager	Manages tasks such as adding, updating, deleting, and displaying.
Category	Classifies tasks into different groups.
FileManager	Handles saving and loading tasks from files.

4. Class Attributes and Behaviours

Class Name	Attributes	Methods (Behaviours)
User	userID : String username : String tm : TaskManager	createTask() viewTasks()
Task	taskID : int taskName : String deadline : String status : String category : String	markComplete() updateTask(String newName, String newDeadline, String newStatus, String newCategory) getDetails() getTaskID() getTaskName() getDeadline() getStatus() getCategory()
TaskManager	taskList : ArrayList<Task> personal : Category work : Category random : Category fm : FileManager	addTask(Task t) addToCategory(Task t) deleteTask(int delete) removeFromCategory(Task t) displayTasks() displayByCategory(int choice) getPendingorCompletedTasks(String statusChoice) updateTask(int taskID, String newName, String newDeadline, String newStatus, String newCategory) saveTasksToFile() markTaskAsComplete(int taskID) getTaskList()
Category	categoryName : String tasks : ArrayList<Task>	addTask(Task t) removeTask(Task t) displayTasks() getCategoryName()
FileManager	fileName : String	saveToFile(ArrayList<Task> taskList) loadFromFile()



3. Sequence Diagram



4. Brief Explanation of the Design

- **Major classes identified and their responsibilities:**

The main classes in the system are User, Task, TaskManager, Category, and FileManager. The User interacts with the system to create and view tasks. The Task class stores task details such as task ID, task name, deadline, and status, and provides operations like marking a task as complete and updating task details. The TaskManager is the central class that manages the collection of tasks by adding, deleting, updating, and displaying them. The Category class is used to group or classify tasks under different categories. The FileManager is responsible for saving task data to a file and loading it back when needed.

- **Relationships used (association / aggregation / inheritance, if any):**

The system mainly uses aggregation, composition, and dependency relationships.

1. There is an aggregation relationship between User and TaskManager, as the User interacts with and uses the TaskManager to perform operations such as creating and viewing tasks.
2. There is a dependency relationship between User and Task, since the User accesses and works with Task objects through the TaskManager.
3. There is an aggregation relationship between Category and Task, because each Category maintains a collection (ArrayList) of Task objects.
4. There is an aggregation relationship between TaskManager and Task, because the TaskManager maintains and manages a collection (ArrayList) of Task objects.
5. There is a composition relationship between TaskManager and Category, since the TaskManager creates and manages Category objects within the system.
6. There is a composition relationship between TaskManager and FileManager, because the TaskManager internally creates and uses the FileManager object for file handling operations.
7. There is a dependency relationship between FileManager and Task, because FileManager temporarily uses Task objects while saving and loading task data from files.

- **Key interaction flow captured in the sequence diagram:**

The sequence diagram shows how the User sends a request to the TaskManager to perform an action such as adding, updating, deleting, or viewing tasks. The TaskManager then interacts with the Task objects to make the required changes. If categorization is needed, the task may also be linked with the Category class. When task data needs to be stored permanently or retrieved later, the TaskManager communicates with the FileManager to save or load the task list. Thus, the sequence diagram captures the flow of control from the user to the manager class and then to the supporting classes.